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\\Research\Investigators - validated on FIT\Frederick\CMRR spectro scans\localizer
TA: 9 sec Coil Selection: Auto Voxel Size: 0.5×0.5×10.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	All Segments
Inline Movie	Off

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	280 mm
FOV Phase	100.0 %
Slice Thickness	10.0 mm
TR	20.0 ms
TE	5.00 ms
Averages	1
Concatenations	3
AutoAlign	---

Contrast - Common

TR	20.0 ms
TE	5.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	40 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	280 mm
FOV Phase	100.0 %
Slice Thickness	10.0 mm
Base Resolution	256
Phase Resolution	50 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve	Off
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	1
Distance Factor	20 %
Position	Isocenter

Geometry - Common

Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	280 mm
FOV Phase	100.0 %
Slice Thickness	10.0 mm
TR	20.0 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	3

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	20.0 ms
Segments	1
Concatenations	3

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	280 mm
FOV Phase	100.0 %
Phase Resolution	50 %

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1

Inline - Subtraction

StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing

Inline Composing	Off

Inline - MapIt

MapIt	None
Flip Angle	40 deg
Measurements	1
Contrasts	1
TE	5.00 ms
TR	20.0 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Fast
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	180 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Acoustic noise reduction	Off

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\Investigators - validated on FIT\Frederick\CMRR spectro scans\localizer_5@center

TA: 40 sec Coil Selection: Auto Voxel Size: 0.9×0.9×5.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	All Segments
Inline Movie	Off

Routine

Slice Group	1
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	20.0 ms
TE	6.00 ms
Averages	1
Concatenations	15
AutoAlign	---

Contrast - Common

TR	20.0 ms
TE	6.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	40 deg

Contrast - Common

Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	256
Phase Resolution	50 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve	Off
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	5
Distance Factor	20 %
Position	Isocenter

Geometry - Common

Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	20.0 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	15

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	20.0 ms
Segments	1
Concatenations	15

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	100.0 %
Phase Resolution	50 %

Physio - PACE

Resp. Control	Off
Concatenations	15

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1

Inline - Subtraction

StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing

Inline Composing	Off

Inline - MapIt

MapIt	None
Flip Angle	40 deg
Measurements	1
Contrasts	1
TE	6.00 ms
TR	20.0 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	260 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Acoustic noise reduction	Off

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\Investigators - validated on FIT\Frederick\CMRR spectro scans\t1_mprage_sag_p2
TA: 3:15 min Coil Selection: Auto Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 2 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	9.1 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2100.0 ms
TE	2.58 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	2100.0 ms
TE	2.58 ms
Magn. Preparation	Non-sel. IR
TI	900 ms
Flip Angle	8 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off
Reordering	Linear

Resolution - Common

FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	80 %
Slice Resolution	80 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	64
Acceleration Factor 3D	1
Reference Lines 3D	24
Phase Partial Fourier	7/8
Slice Partial Fourier	Off
Asymmetric Echo	Allowed
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	Prescan
Image Filter	On

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	9.1 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2100.0 ms
Multi-Slice Mode	Single Shot
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	2100.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	Non-sel. IR
TI	900 ms
Dark Blood	Off
FOV Read	250 mm
FOV Phase	100.0 %
Phase Resolution	80 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Inline - MapIt

MapIt	None
Flip Angle	8 deg
Measurements	1
Contrasts	1
TE	2.58 ms
TR	2100.0 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	tfl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear

Sequence - Part 1

Bandwidth	150 Hz/Px
Echo Spacing	8.84 ms
Asymmetric Echo	Allowed
Turbo Factor	154

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	On

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators - validated on FIT\Frederick\CMRR spectro scans\ep2d_bold_s8_2mm

TA: 12:42 min Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm^3 Acc:: 8 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	64
Distance Factor	0 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	627.0 ms
TE	30.00 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	627.0 ms
TE	30.00 ms
MTC	Off
Flip Angle	60 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1200
Delay in TR	0.00 ms

Resolution - Common

FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
Base Resolution	100

Resolution - Common

Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	EPI/Separate
Acceleration Factor PE	1
SMS Factor	8
Phase Partial Fourier	7/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	64
Distance Factor	0 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	627.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	200 mm
R >> L	200 mm
F >> H	128 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	627.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0

BOLD

Model Transition States	Off
Temp. Highpass Filter	Off
Threshold	4.00
Paradigm Size	40
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Baseline
Meas[12]	Baseline
Meas[13]	Baseline
Meas[14]	Baseline
Meas[15]	Baseline
Meas[16]	Baseline
Meas[17]	Baseline
Meas[18]	Baseline
Meas[19]	Baseline
Meas[20]	Baseline
Meas[21]	Active
Meas[22]	Active
Meas[23]	Active
Meas[24]	Active
Meas[25]	Active
Meas[26]	Active
Meas[27]	Active
Meas[28]	Active
Meas[29]	Active
Meas[30]	Active
Meas[31]	Active
Meas[32]	Active
Meas[33]	Active
Meas[34]	Active
Meas[35]	Active
Meas[36]	Active
Meas[37]	Active
Meas[38]	Active
Meas[39]	Active
Meas[40]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	1200
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Performance

Sequence - Part 1

Bandwidth	2174 Hz/Px
Echo Spacing	0.62 ms
Free Echo Spacing	Off
EPI Factor	100

Sequence - Part 2

Introduction	On

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators - validated on FIT\Frederick\CMRR spectro scans\svs_se_30
TA: 2:50 min Coil Selection: Auto VoI: 20×20×20 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
VoI A >> P	20 mm
VoI R >> L	20 mm
VoI F >> H	20 mm
TR	2000.0 ms
TE	30.00 ms
Averages	80

Contrast - Common

TR	2000.0 ms
TE	30.00 ms
Flip Angle	90 deg
Preparation Scans	4
Averages	80
Water Suppression	Water Saturation
Water Suppr. BW	50 Hz
Spectral Suppr.	None

Resolution - Common

Vector Size	1024
Normalize	Prescan

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
VoI A >> P	20 mm
VoI R >> L	20 mm
VoI F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---

Geometry - AutoAlign

Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	When Changed
Assume Silicone	Off
Adj. Water Suppr.	On

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	2000.0 ms

Physio - PACE

Resp. Control	Off

Sequence - Common

Sequence Name	svs_se
Preparation Scans	4
Delta Frequency	-2.30 ppm
Ref. Scan Mode	Inline Correction
No. of Ref. scans	1
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	1200 Hz
Acquisition Duration	853 ms
Remove Oversampling	On
Freq. Corr. Accumulation	Off
Incr. Gradient Spoiling	Off

\\Research\Investigators - validated on FIT\Frederick\CMRR spectro scans\eja_svs_press
TA: 3:01 min Coil Selection: Auto VoI: 20×20×20 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
VoI A >> P	20 mm
VoI R >> L	20 mm
VoI F >> H	20 mm
TR	2000.0 ms
TE	30.00 ms
Averages	80

Contrast - Common

TR	2000.0 ms
TE	30.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	80
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	80.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	1024
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
VoI A >> P	20 mm
VoI R >> L	20 mm
VoI F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	2000.0 ms

Sequence - Common

Sequence Name	press

Sequence - Common

Preparation Scans	4
Delta Frequency	-2.30 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	1200 Hz
Acquisition Duration	853 ms
Remove Oversampling	On

Sequence - Special

Excite pulse duration	2600 us
Refocus pulse duration	5200 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	3.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
Forced min. TE1	0 ms
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	90 deg
OVS flip angle SL	90 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Resolve averages	Off
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
'RR' refoc. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None

\\Research\Investigators - validated on FIT\Frederick\CMRR spectro scans\eja_svs_mpress_ws_rACC
TA: 42 sec Coil Selection: Auto Vol: 20×35×25 mm^3 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	R3.1 A71.3 H32.2 mm
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	35 mm
Vol F >> H	25 mm
TR	3000.0 ms
TE	46.64 ms
Averages	1

Contrast - Common

TR	3000.0 ms
TE	46.64 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	2
Averages	1
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	R3.1 A71.3 H32.2 mm
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	35 mm
Vol F >> H	25 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	R3.1 A71.3 H32.2
R	3.1 mm
A	71.3 mm
H	32.2 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	On
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	R3.1 A71.3 H32.2 mm
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	35 mm
F >> H	25 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	mpres

Sequence - Common

Preparation Scans	2
Delta Frequency	-1.70 ppm
Measurements	9
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Special

Excite pulse duration	2600 us
Refocus pulse duration	5200 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	1.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
Forced min. TE1	0 ms
MEGA flip angle	180 deg
VAPOR flip angle	100 deg
VAPOR delay 8	16 ms
VAPOR delay 7	60 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	90 deg
OVS flip angle SL	90 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Use expt. SAR calc	Off
Weighted averages	Off
Check RF clipping	Off
Send ref. scans	Off
PRESS+4	Off
Symm. MEGA timing	Off
MEGA water suppr.	Off
Inversion pulse	Off
'RR' refoc. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	WS FA
WS FA inc.	10 deg
Measurements	9

Sequence - Special

Number of label freqs.	1
Editing pulse freq. [1]	1.90 ppm
Editing pulse BW	100.00 Hz

\\Research\Investigators - validated on FIT\Frederick\CMRR spectro scans\eja_svs_mpress_Am25D1200 bw54
TA: 10:02 min Coil Selection: Auto Vol: 20×35×25 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	R3.1 A71.3 H32.2 mm
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	35 mm
Vol F >> H	25 mm
TR	3000.0 ms
TE	68.00 ms
Averages	96

Contrast - Common

TR	3000.0 ms
TE	68.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	2
Averages	96
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	110.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	R3.1 A71.3 H32.2 mm
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	35 mm
Vol F >> H	25 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	R3.1 A71.3 H32.2
R	3.1 mm
A	71.3 mm
H	32.2 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	On
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	R3.1 A71.3 H32.2 mm
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	35 mm
F >> H	25 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	mpres
Preparation Scans	2
Delta Frequency	-1.70 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	1200 Hz
Acquisition Duration	1706 ms
Remove Oversampling	On

Sequence - Special

Excite pulse duration	2600 us
Refocus pulse duration	5200 us
Spoiler max. amplitude	25.00 mT/m
Spoiler amp. ratio	1.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1200 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	240 us
Forced min. TE1	0 ms
MEGA flip angle	180 deg
VAPOR flip angle	150 deg
VAPOR delay 8	16 ms
VAPOR delay 7	65 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	90 deg
OVS flip angle SL	90 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Resolve averages	On
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
PRESS+4	Off
Symm. MEGA timing	Off
MEGA water suppr.	On
Inversion pulse	Off
'RR' refoc. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None
Number of label freqs.	2

Sequence - Special

Editing pulse freq. [1]	7.50 ppm
Editing pulse freq. [2]	1.90 ppm
Editing pulse BW	54.00 Hz

\\Research\Investigators - validated on FIT\Frederick\CMRR spectro scans\eja_svs_mpress_rACC_EC
TA: 23 sec Coil Selection: Auto Vol: 20×35×25 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	R3.1 A71.3 H32.2 mm
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	35 mm
Vol F >> H	25 mm
TR	3000.0 ms
TE	68.00 ms
Averages	1

Contrast - Common

TR	3000.0 ms
TE	68.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	2
Averages	1
VAPOR	RF Off
VAPOR suppr.	Water Suppr.
Water s. BW	110.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	R3.1 A71.3 H32.2 mm
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	35 mm
Vol F >> H	25 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	R3.1 A71.3 H32.2
R	3.1 mm
A	71.3 mm
H	32.2 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	On
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	R3.1 A71.3 H32.2 mm
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	35 mm
F >> H	25 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	mpres

Sequence - Common

Preparation Scans	2
Delta Frequency	-1.70 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	1200 Hz
Acquisition Duration	1706 ms
Remove Oversampling	On

Sequence - Special

Excite pulse duration	2600 us
Refocus pulse duration	5200 us
Spoiler max. amplitude	25.00 mT/m
Spoiler amp. ratio	1.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1200 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	240 us
Forced min. TE1	0 ms
MEGA flip angle	180 deg
VAPOR flip angle	130 deg
VAPOR delay 8	16 ms
VAPOR delay 7	60 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	90 deg
OVS flip angle SL	90 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
PRESS+4	Off
Symm. MEGA timing	Off
Inversion pulse	Off
'RR' refoc. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None
Number of label freqs.	2
Editing pulse freq. [1]	7.50 ppm
Editing pulse freq. [2]	1.90 ppm
Editing pulse BW	54.00 Hz

\\Research\Investigators - validated on FIT\Frederick\CMRR spectro scans\svs_edit
TA: 6:32 min Coil Selection: Auto Vol: 20×20×20 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm
TR	2000.0 ms
TE	68.00 ms
Averages	96

Contrast - Common

TR	2000.0 ms
TE	68.00 ms
Flip Angle	90 deg
Preparation Scans	4
Averages	96
Water Suppression	Weak Water Suppression
Water Suppr. BW	35 Hz
Edit Frequency	1.90 ppm
Edit Bandwidth	50 Hz
Edit Mirror Frequency	4.70 ppm

Resolution - Common

Vector Size	1024
Normalize	Prescan

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	When Changed
Assume Silicone	Off
Adj. Water Suppr.	On

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Sequence - Common

Sequence Name	svs_edit
Preparation Scans	4
Delta Frequency	-2.80 ppm
Ref. Scan Mode	Off
Phase Cycling	Auto

Sequence - Common

Bandwidth	2000 Hz
Acquisition Duration	512 ms
Remove Oversampling	On
Freq. Corr. Accumulation	On
Incr. Gradient Spoiling	Off

\\Research\Investigators - validated on FIT\Frederick\CMRR spectro scans\eja_svs_mpress

TA: 6:46 min Coil Selection: Auto Vol: 20×20×20 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm
TR	2000.0 ms
TE	68.00 ms
Averages	96

Contrast - Common

TR	2000.0 ms
TE	68.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	96
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	2000.0 ms

Sequence - Common

Sequence Name	mpres

Sequence - Common

Preparation Scans	4
Delta Frequency	-2.80 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Special

Excite pulse duration	2600 us
Refocus pulse duration	5200 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	1.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
Forced min. TE1	0 ms
MEGA flip angle	180 deg
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	90 deg
OVS flip angle SL	90 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Resolve averages	Off
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
PRESS+4	Off
Symm. MEGA timing	On
MEGA water suppr.	Off
Inversion pulse	Off
'RR' refoc. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None
Number of label freqs.	2
Editing pulse freq. [1]	9.00 ppm

Sequence - Special

Editing pulse freq. [2]	1.90 ppm
Editing pulse BW	100.00 Hz

\\Research\Investigators - validated on FIT\Frederick\CMRR spectro scans\eja_svs_mslaser
TA: 6:51 min Coil Selection: Auto VoI: 20×20×20 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
VoI A >> P	20 mm
VoI R >> L	20 mm
VoI F >> H	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	64

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	64
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
VoI A >> P	20 mm
VoI R >> L	20 mm
VoI F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	mslsr

Sequence - Common

Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Special

Excite pulse duration	8960 us
Refocus pulse duration	8960 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	3.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
MEGA flip angle	180 deg
HS refoc. pulse N	1
HS refoc. pulse R	20
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	0 deg
OVS flip angle SL	0 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Resolve averages	Off
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
MEGA water suppr.	Off
Inversion pulse	Off
GOIA refoc. pulses	Off
Asym. excit. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None
Number of label freqs.	2
Editing pulse freq. [1]	9.00 ppm

Sequence - Special

Editing pulse freq. [2]	1.90 ppm
Editing pulse BW	100.00 Hz

\\Research\Investigators - validated on FIT\Frederick\CMRR spectro scans\eja_svs_press
TA: 3:39 min Coil Selection: Auto Vol: 20×20×20 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	64

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	64
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	press

Sequence - Common

Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Special

Excite pulse duration	2600 us
Refocus pulse duration	5200 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	3.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
Forced min. TE1	0 ms
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	90 deg
OVS flip angle SL	90 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Resolve averages	Off
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
'RR' refoc. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None

\\Research\Investigators - validated on FIT\Frederick\CMRR spectro scans\eja_svs_steam
TA: 3:39 min Coil Selection: Auto Vol: 20×20×20 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	64

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
TM	75.00 ms
Flip Angle	90 deg
Preparation Scans	4
Averages	64
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	steam

Sequence - Common

Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Special

RF pulse duration	2560 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	1.50
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	90 deg
OVS flip angle SL	90 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Resolve averages	Off
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
Symmetric RF pulses	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None

\\Research\Investigators - validated on FIT\Frederick\CMRR spectro scans\eja_svs_slaser
TA: 3:39 min Coil Selection: Auto Vol: 20×20×20 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	64

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	64
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	slasr

Sequence - Common

Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Special

Excite pulse duration	8960 us
Refocus pulse duration	8960 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	3.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
HS refoc. pulse N	1
HS refoc. pulse R	20
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	0 deg
OVS flip angle SL	0 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Resolve averages	Off
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
GOIA refoc. pulses	Off
Asym. excit. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None